Available online at www.sciencedirect.com
SCIENCE DIRECT®
Brain and Language 94 (2005) 19–31
Brain and Language
www.elsevier.com/locate/b&l

The Boston Naming Test in Swedish: Normative data

I.M. Tallberg*

Division of Logopedics and Phoniatrics, Department of Clinical Sciences, Karolinska Insitutet, B69, Karolinska University Hospital, S 141 86 Stockholm, Sweden

Accepted 10 November 2004
Available online 7 January 2005

Abstract

The purpose of the present study was to introduce a Swedish version of the Boston Naming Test and to offer normative data based on a sample of native Swedish-speaking healthy adults stratified concerning age, gender, and length of education. The subjects were assessed with other lexical tests and half of the group also performed tests of global cognitive function. A semantic analysis of the responses was performed and the typical Swedish naming of the pictures of BNT was investigated. The results showed that long education, high performance on tests of global cognitive ability (FSIQ), verbal fluency, and other lexical tests had a significant positive association to a good performance on BNT whereas age and gender had a minor influence. The response analysis showed that the more difficult an item was, the more variations in the response pattern. To conclude, the Boston Naming Test is appropriate for use in a Swedish-speaking context and an effective test for assessment of naming ability.
© 2004 Elsevier Inc. All rights reserved.

Keywords: Boston Naming Test; Normative data; Swedish native-speaking sample; Response analysis

1. Introduction

Confrontation picture naming tests have always been widely used in aphasiology and cognitive research because they are easily administered and sensitive to language disturbances (Rubin et al., 1998). The probably most distributed naming test around the world is the Boston Naming Test (BNT), introduced in 1983 by Kaplan, Goodglass and Weintraub. BNT is used in clinical work as well as in research and the knowledge of how people from different groups and contexts perform on the test is deep and extensive. The vast source of knowledge, emanating from clinical and experimental research using BNT, is a strong argument for the choice to use BNT. Adaptations and collection of normative data from new populations and languages, like Swedish in the present study, are still needed.

There is no published version of BNT for native Swedish-speaking persons, although it has been used in Sweden in research (for example, Almkvist, Bäckman, Basun, & Wahlund, 1993) and in clinical practice. Many different translations have been used, which is not satisfactory. When preparing for a Swedish version several aspects were considered. Beside a proper translation and adaptation based on knowledge of a normal and correct way to name the pictures, normative data from a stratified population is necessary. In normative studies of BNT, demographic factors usually are well known and a stratified selection has been made regarding age, gender, and education. Usually some form of health assessment is performed in the selection, either by use of a general practitioner's examination, a health questionnaire or alike. Sometimes also cognitive and language ability is known by inclusion of different tests.

The adaptation of picture confrontation tests to populations with different cultures and languages requires special considerations. First, the selection of pictures

* Fax: +46 8 58581505.
E-mail address: ing-mari.tallberg@klinvet.ki.se.

0093-934X/$ - see front matter © 2004 Elsevier Inc. All rights reserved.
doi:10.1016/j.bandl.2004.11.004

might be very inappropriate for the new population. Second, the linguistic features of the target words might be impossible to compare. The problem of transforming BNT for English-speaking subjects into another language and culture was encountered in a study aiming to introducing the Korean version of the Boston Naming Test (Kim & Na, 1999). In their study, the original BNT was the point of departure but most of the pictures had to be altered due to cultural differences. Unfortunately, a substantial altering of pictures makes a cross-linguistic comparison inappropriate. The linguistic differences can also be assumed to be so salient that this has a substantial impact on the results. The word form can differ essentially between languages according to phonological complexity, word length and alike. In a study of a Dutch-speaking Belgian population, it was found that linguistic features had no impact on the overall BNT-score (Mariën, Mampaey, Vervaet, Saerens, & De Deyn, 1998). The Dutch language has a linguistic relation to English and is assumed to cause relatively minor problems in comparison with languages with a very weak relation to English like for example, Finnish or Korean. The translation from English and the understanding about what the pictures of the test are supposed to show are other aspects that demand careful consideration. In the Dutch study, standard dictionaries were used for the translations and the decision of which translations to use was made in consensus. Knowledge of how the normal population typically names the sometimes unfamiliar objects can be of further help in the decisions of which responses that should be regarded as correct. A response analysis can be of value in the adaptive process.

Normative data are available for many populations. Reports are published concerning healthy adults varying in gender, age, and education (for example: Neils et al., 1995; Tombaugh & Hubley, 1997; Van Gorp, Satz, Kiersch, & Henry, 1986; Welch, Doineau, Johnson, & King, 1996). There is a general agreement concerning the overall effect of age, education and gender on BNT performance. The naming ability decreases with higher age and lower education and men perform somewhat better than women do. Neils et al. (1995) found that the educational background had a greater impact than age but that this impact was lower in very high age. The knowledge of the interactive effect between age and education on naming was confirmed and deepened by for example Welch et al. (1996) and by Tombaugh and Hubley (1997). Other reports concern diseased or brain-damaged groups of adults for example groups with dementia (Henderson, Mack, Freed, Kempler, & Andersen, 1990; Margolin, Pate, Friedrich, & Elia, 1990; Nicholas, Obler, Au, & Albert, 1996). Also studies of bilingual populations (Kohnert, Hernandez, & Bates, 1998) have been performed.

A confrontation-naming test is an example of a test that is sensitive to brain damage (Rubin et al., 1998). Consequently, BNT is often used as one of many tests to evaluate cognition in cases with suspected cognitive impairment and possible anomia but with no dysphasia in the strict sense. Other aspects of lexical ability are more resistant to influence of disturbed brain function and can be assessed by means of so called "hold"-tests like lexical decision tests and pronunciation tests like NART [National Adult Pronunciation test; Nelson and O'Connell (1978)]. A knowledge of how results on naming tests relate to results on these so called hold-tests could be valuable in future research and in clinical evaluation of patients with suspected brain dysfunction.

Response analysis based on BNT has been used in investigations of the lexico-semantic network both in normal and clinical populations. The research has been focused on the distribution of errors and choice of words (Tombaugh & Hubley, 1997) or on aspects of the search process evaluated by measures of for example response time or lexical decision latency (Moberg, Ferraro, & Petros, 2000). Nicholas et al. (1996) studied closeness to BNT target words in Alzheimer patients and healthy controls but they could not find any expected differences between groups. Others, like Hodges, Salmon, and Butters (1991), found qualitative differences concerning erroneous response pattern in AD and controls. Controls used descriptive utterances, while the AD patients made semantic errors. In a study with the main purpose to obtain normative data for a native Dutch-speaking Belgian population, an analysis of the erroneous responses was performed. The incidence of different error types concerning word forms and choices as well as circumlocutions and neologisms was studied (Mariën et al., 1998). The knowledge of the normal erroneous response pattern will be of value in the clinical examination of Dutch-speaking patients having language disturbances. It would be of value to know more about how a healthy adult Swedish population responds and substitutes for lack of semantic and/or lexical knowledge as a basis for further research on clinical populations.

Furthermore, problems with BNT scoring will be addressed in the present study. Lopez, Arias, Hunter, and Scott (2003) have investigated the scoring process in BNT, and they have found that the instructions for the BNT were insufficient which caused large differences in scoring when different scoring methods were compared. Arbitrariness in decision of which response alternative that should be evaluated as correct and a poor knowledge of how the pictures actually are named by normal adults are issues that will be investigated in the present article. In the original American version of BNT, the target is available in the protocol but the method used for selection of target responses is not fully specified.

The primary purpose of the present study was to introduce a Swedish version of BNT that is comparable to the English original version of BNT and to offer normative data based on a sample of the Swedish

population. Another aim was to investigate the responses from a lexico-semantic point of view. The tasks were accomplished by an examination of a group of 111 healthy Swedish-speaking adults. First, the adaptation to Swedish had to be consistent with the actual choices of lexical terms made by the healthy population. This procedure made it possible to suggest Swedish scoring rules and norms. Second, the effect of gender, age, education, and general cognition on BNT, as measured by number of correct responses was examined. Third, an analysis of response types was performed in relation to background data on age, gender, education, as well as general cognitive ability.

2. Method

2.1. Subjects

A healthy native Swedish sample consisting of 111 individuals (61 females and 50 males) was included in the present study. The participants were recruited from different work places, universities, and organizations. The participants all lived in central parts of Sweden, an area without marked dialectal features. A specification of the dialect of each subject was however, not performed. The selection was stratified and an even distribution of gender (female/male), age (30 years or less/31–65 years/>65 years) and levels of education (12 years or less/>12 years) was sought after. The range of age was between 21 and 80 years with an average of 49.5. The level of education was in average 12.8 years and ranged between 6 and 22 years. All individuals who accepted to participate in the study also agreed to fill in a health status questionnaire (Sullivan & Karlsson, 1998). Any individual with a history of psychiatric or neurological diseases or with head trauma, abuse or other illnesses that were assumed to have an influence on the performances were excluded. All participants gave their informed written consent. The local Ethics Committee of Huddinge University Hospital (Dnrs 394/02, 400/02, 442/02) approved the study (see Table 1).

2.2. Procedure

All 111 subjects were administrated the 60-item Boston Naming Test and the same standardized instructions were given. We chose not to use the standard procedure with respect to phonological and semantic cueing. Phonological cueing was not attempted because we had no Swedish version to base the cueing upon. A semantic cueing could have been used in cases of suspected visual interpretation problems but we decided not to. Tombaugh and Hubley (1997) used semantic prompting both in cases of suspected visual interpretation problems and when the person did not know the answer but despite this generous use of prompting the effect was minor. Further, it was also of interest to study to which degree visual associations occurred in a healthy population. The examiners made notations of all the responses on a protocol and the test sessions were audio taped on a tape recorder. The examiners were students in their final semester of a Master's programme in speech language pathology and they got special training to perform the testing in a proper and uniform way. Most of the subjects (70) was also administrated the Controlled Oral Word Association test, (COWAT, Benton & Hamshire, 1989) using the letters F, A, S, and 48 subjects performed the semantic category variant with the semantic category "animals" as objective (ref). A short version of WAIS-R (Wechsler, 1981) was administered to 54 participants Finally, 62 subjects performed a lexical decision test (Almkvist, Adveen, Henning, & Tallberg, in preparation) and 48 subjects a pronunciation test of irregularly pronounced words (Tallberg, Wenneborg, & Almkvist, in preparation). The total test time varied according to the amount of tests used and the tests were administrated in one or two sessions.

Table 1
Demographic and background characteristics for all subjects

N (females/males)	111 (61/50)
Age ($M \pm SD$), years	49.5 ± 19.4
Education ($M \pm SD$), years	12.8 ± 3.3
Global cognition ($M \pm SD$), FSIQ score	109.7 ± 17.3

2.3. The development of a Swedish version of the Boston Naming Test

The analysis and adaptation was made in the following steps:

All responses given within a 20 s limit were noted in full length. Responses consisting of a couple or more alternatives occurred but only the subjects' final word choice was selected for the analysis. A preliminary protocol was used, one among several versions used in Sweden for BNT. All responses in accordance with this preliminary version were noted. When all the 111 subjects used the same Swedish word for the item, it was decided that this word should be the Swedish target-word. For example "Item 1. (bed) gave no other responses than the Swedish "*säng*"(= *bed*).

The following procedure was then followed:

- The English target words were translated by using a standard lexicon (Nordstedts Comprehensive English–Swedish Dictionary, 1993).
- Alternative responses were checked in Swedish standard lexicons: SvenskaAkademins ordlista (SAOL, 1998), Synonym-lexicon (Norstedts, 1992), and the Swedish National Encyclopaedia (NE, 1997).
- All alternative responses (363 items) were classified into different lexical and semantic categories.

The categories were defined with respect to semantic closeness to the target and to hierarchical organization within the network.

2.4. Classification of responses

The following categories were used:

1. In accordance with final version target word
2. No response, or aborted response ("*ste*…" for *stethoscope*)
3. Phonologically incorrect response ("*sfnix*" for Swedish *sfinx* (*sphinx*))
4. Subordinated word ("*cottage*" for *house*)
5. Superordinated word ("*animal*" for *beaver*)
6. Sideordinated word ("*rat*" for *beaver*)
7. Synonym according to lexicon

8a. Visual association ("*tape*" for *whistle*)
8b. Semantic association ("*geyser*" for *volcano*)
8c. Contextual association ("*Egypt*" for *pyramid*)
8d. Phonological association ("*glove*" for *globe*)

9. Unrelated

10a, b, c, d. Descriptive utterances.

The descriptive utterances could not be found in Swedish lexicons and were classified as substitutions for the target word. The classification of the descriptive utterances was based on degree of specificity and ranged from exactly and close (10a) to over generalized and empty (10d). A specific utterance (10a) contained at least two specific and relevant attributive words, for example "*Eskimo-hut*" for *igloo*. An example of a semantically irrelevant and unspecific utterance (10d) was "*a drawing thing*" for *tripod*. Use of a word (or a portmanteau word like "*Eskimo-hut*") or a complete clause was not the determining factor for the categorization but instead the relevance and specificity of the attributive words that the utterances contained.

To examine the reliability of the method for classification the total interrater agreement between the first classificator (the author) and a second trained rater was calculated. The second rater was blind to the subjects' identity and test results. The result was in exact agreement in 83% of the 363 items, which is regarded as good.

2.5. Swedish target words

The final Swedish version was developed in two steps. *In the first step* only those responses that were exactly in accordance with the first preliminary version and also were in accordance with definitions and translations according to Swedish standard lexicons were scored as correct. In the *second step* words that were decided to be synonyms according to lexicon and also all subordinated words were allowed and scored as correct. Most of the subordinated words shared the same word stem with the target word (for example, "*tennis-racquet*" instead of only *racquet*). When possible, the properties of the words in the Swedish version should be comparable with and at the same semantic level as in the English version. Sometimes it was necessary to score a word at another semantic level than the English version word as correct (see for example item 27 *globe*, which corresponds to the Swedish word *jordglob* that is subordinated to the Swedish word *globe*). Some monomorphemic English items were translated into compound Swedish words (7 items) but they did not differ much with respect to number of syllables or phonological complexity. The morphological structure of the translated words was therefore considered to be of minor importance. Another intention was that the mean score of correct responses should be at the same level as the English version.

The final Swedish version was decided in consensus with two experienced speech pathologists. It was decided that a Swedish manual for scoring should include comments on problematic scoring issues. See the Appendix for the final version.

2.6. Assessment of cognitive function and lexical ability

Global cognitive function was evaluated by means of five subtests (Information, Digit Span, Similarities, Block Design, and Digit Symbol) from Wechsler Adult Intelligence Scale-Revised (WAIS-R, Wechsler, 1981). In addition, four tests of lexical knowledge was used focusing on different aspects of lexical ability, (1) two tests of word retrieval and fluency (Letter fluency and Category fluency), and (2) test of the passive lexicon by assessment of the ability to separate between words and nonwords (SLDT; Swedish Lexical Decision test; Almkvist et al., in preparation) and a pronunciation test of irregularly pronounced words (IPW; Irregularly Pronounced Words Pronunciation test; Tallberg et al., in preparation). The two latter (SLDT and IPW) are so called "hold tests" regarded to be relatively resistant against brain damage (Lezak, 1995).

2.7. Statistical analysis

The statistical calculations were performed by means of SPSS for Windows (Statistical Package for the Social Sciences, 1999). Demographic data were presented as means and standard deviations ($M \pm SD$). Relationship between BNT variables and demographic as well as indices of cognitive function were calculated using Pearson's correlation coefficient or biserial correlation coefficients when dichotomous variables were used. Statistical significance of group differences was evaluated by means of Student's t test. A principal component analysis was employed on all subjects to study the pattern of relationship between Letter and Category fluency, as well as SLDT, IPW, and BNT using eigen values >1 as criteria for extraction of number of factors and varimax rotation of the component matrix.

3. Results

3.1. Adaptation of the original English version of BNT to a Swedish version of BNT

The first step was to investigate the spontaneous responses to BNT made by native Swedish-speaking adults. The responses for each item was categorized according to the 16 response categories as described above in Section 2. The frequency of all responses for each category across all 60 BNT pictures is presented in Table 2. The target responses of the preliminary version of the Swedish BNT were used in 88.4% of all responses. The mean score and standard deviation for all individuals was 47.58 ± 4.15 which is a low mean score when compared with similar studies (Kaplan, Goodglass, & Weintraub, 1983).

In the next step, it was decided to accept responses in total agreement, synonyms and subordinated words as correct responses in the revised version of a Swedish BNT. The preliminary list of target words was changed in a few cases, to adapt the first version to correct responses that really were used. The most common response given by the native Swedish-speaking individuals was defined as the correct response for the revised Swedish BNT. A final version of target words was constructed as a result of this procedure see the Appendix. When synonyms and subordinated words were included as correct responses these words were used in 97.4% of all responses. This procedure was possible to follow in a strict way, which was desired to avoid arbitrariness, and to make possible a future standardized scoring. Consequently, the scoring as correct of subordinated words was questionable in a few issues (*dromedary* for *camel*). The mean score and standard deviation of correct responses for the final Swedish version of BNT and for all participating individuals was 53.04 ± 4.18.

For the final version of the Swedish BNT, the distribution of number of correct responses fitted to a normal frequency distribution. According to χ^2-test, the hypothesis could not be rejected ($\chi^2 = 15.2$; $df = 11$; $p > .10$), see Fig. 1.

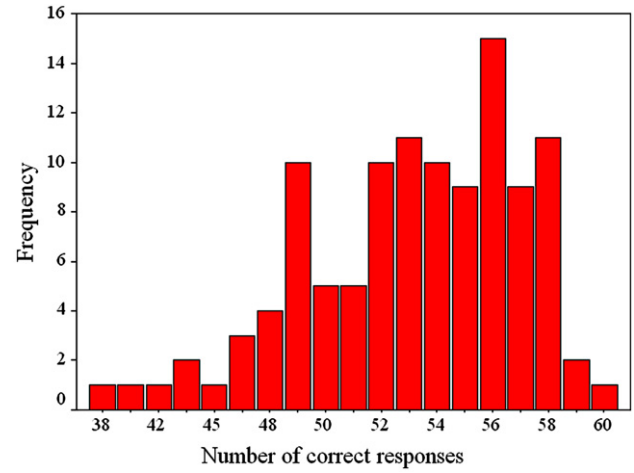

Fig. 1. Frequency distribution of number of correct responses to the Swedish BNT.

3.2. Correct responses in relation to demographic and cognitive characteristics

Difficulty in naming was analyzed in two ways, ability to name correctly and frequency of "omitted" responses. The 5 items that were named correctly in less than 75% of the total amount of responses all appeared among items 40–60. The most difficult item was *latch* (totally 19/111 correct responses), followed by *knocker* (39/111 correct responses), *scroll* (56/111 correct responses), and *muzzle* and *unicorn* (67/111 correct responses each). The items that resulted in the highest frequency of "don't know" responses were *unicorn*, *compass*, *sphinx*, *yoke*, and *trellis*. The final order of items of the Swedish BNT based on frequency of a correct response is presented in the Appendix. This sequence differed somewhat from the sequence of the original English version of BNT. The result showed that no items had to be replaced due to cultural differences.

3.2.1. Number of correct responses

Number of years in formal education was significantly associated with the number of correct responses of the final version of Swedish BNT as presented in Table 3. Age and gender was not significantly associated with number of correct responses. For three items only out of all 60 items, there was a minor effect of gender (in favour of men for *compass* and *scroll*; in favour of women for *asparagus*). Furthermore, global cognitive ability (FSIQ) was significantly correlated with total number of correct responses. The relationship with global cognition was mainly due to the strong associa-

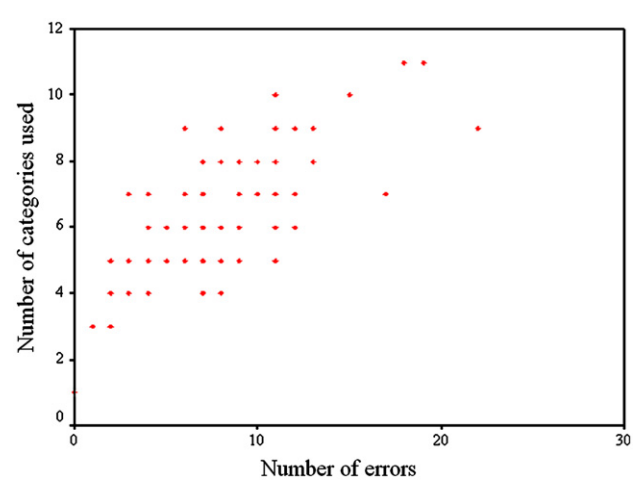

Fig. 2. Scatter plot depicting the total number of errors and the number of various categories used in erroneous responses for a representative sample of healthy native Swedish individuals.

Table 2
Frequency of type of response to items 1–60 of the Boston Naming Test

Item	Responses	Response category															
		Target	Synonym	Sub-ordinated	Phonologic error	Super-ordinated	Side-ordinated	Visual association	Semantic association	Contextual association	Phonologic association	Unrelated	Very specific	Specific	Less specific	Unspecific	Omitted
		1	7	4	3	5	6	8a	8b	8c	8d	9	10a	10b	10c	10d	2
1. bed	111	111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2. tree	111	105	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0
3. pencil	111	88	0	23	0	0	0	0	0	0	0	0	0	0	0	0	0
4. house	111	107	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0
5. whistle	111	109	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0
6. scissors	111	111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7. comb	111	111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8. flower	111	105	0	5	0	0	0	0	0	0	0	0	0	1	0	0	0
9. saw	111	92	0	19	0	0	0	0	0	0	0	0	0	0	0	0	0
10. toothbr.	111	110	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
11. helico.	111	111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12. broom	111	62	1	46	0	1	1	0	0	0	0	0	0	0	0	0	0
13. octopus	111	103	0	0	0	0	3	3	0	0	0	0	0	0	0	0	2
14. mushroom	111	107	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0
15. hanger	111	89	13	9	0	0	0	0	0	0	0	0	0	0	0	0	0
16. wheelch.	111	110	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
17. camel	111	101	0	10	0	0	0	0	0	0	0	0	0	0	0	0	0
18. mask	111	93	0	17	0	0	0	0	0	1	0	0	0	0	0	0	0
19. pretzel	111	105	0	5	0	0	1	0	0	0	0	0	0	0	0	0	0
20. bench	111	57	1	46	0	0	6	0	0	0	0	0	0	1	0	0	0
21. racquet	111	55	0	56	0	0	0	0	0	0	0	0	0	0	0	0	0
22. snail	111	111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
23. volcano	111	98	0	1	0	0	0	2	1	9	0	0	0	0	0	0	0
24. seahorse	111	93	0	0	0	1	10	0	1	0	0	0	0	0	1	0	5
25. dart	111	83	0	27	0	0	0	0	0	0	0	0	0	0	0	1	0
26. canoe	111	80	0	26	0	2	3	0	0	0	0	0	0	0	0	0	0
27. globe	111	107	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0
28. wreath	111	106	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0
29. beaver	111	100	0	0	0	4	6	0	0	0	0	0	0	0	0	0	1
30. harmonica.	111	111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
31. rhinoche.	111	110	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
32. acorn	111	98	0	0	0	6	4	2	0	0	0	0	0	0	0	0	1
33. igloo	111	106	0	0	0	1	3	0	0	0	0	0	1	0	0	0	0
34. stilts	111	106	0	0	0	1	2	0	0	0	0	0	0	1	0	0	1
35. dominoes	111	44	0	52	0	4	9	0	0	0	0	0	0	0	0	0	2
36. cactus	111	107	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2
37. escalator	111	111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
38. harp	111	98	0	0	0	1	10	0	0	0	0	0	0	0	0	0	2
39. hammock	111	106	0	0	0	0	0	0	0	0	0	0	0	0	1	0	4
40. knocker	111	1	0	38	0	0	9	3	16	1	31	0	3	3	2	0	4
41. pelic	111	93	0	0	0	8	10	0	0	0	0	0	0	0	0	0	1
42. stetosc.	111	101	0	0	0	0	0	0	1	0	0	0	0	0	0	0	9
43. pyra	111	108	0	1	0	0	0	1	0	0	0	0	0	0	0	0	1
44. muzzle	111	66	1	0	0	0	20	0	1	1	0	0	0	6	6	2	8
45. unicorn	111	66	1	0	0	1	7	0	15	0	0	0	0	0	4	1	16
46. funnel	111	110	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
47. accordion	111	109	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0

48. noose	111	59	0	21	0	18	0	0	0	3	0	0	0	1	3	5	0	1
49. asparagus	111	86	0	0	0	3	2	12	0	0	0	0	0	0	0	1	0	7
50. compass	111	86	0	0	0	0	0	1	0	0	0	0	0	0	1	2	11	20
51. latch	111	15	0	4	0	17	45	0	3	0	7	0	0	0	4	9	0	7
52. tripod	111	68	0	6	0	1	15	0	4	0	0	0	0	0	4	1	2	10
53. scroll	111	12	0	44	0	1	2	0	24	11	0	0	0	2	3	1	0	11
54. tongs	111	31	0	63	0	3	5	0	0	2	0	0	0	1	3	1	1	1
55. sphinx	111	83	0	0	5	3	0	0	0	2	0	0	0	0	1	1	2	14
56. yoke	111	65	0	9	0	1	7	0	2	1	0	0	0	0	0	4	4	18
57. trellis	111	89	0	1	0	0	0	0	2	0	0	0	0	1	2	3	0	13
58. palette	111	85	0	4	2	0	2	0	2	2	5	0	0	0	0	1	1	7
59. protractor	111	79	0	0	0	1	10	2	1	3	0	0	0	1	3	2	1	8
60. abacus	111	76	9	0	0	3	0	0	2	0	0	0	0	3	1	5	1	11
Sum resp.	6660	5887	26	573	7	87	194	28	75	36	43	0	0	13	37	50	17	187
%	100	88.4	0.4	8.6	0.1	1.3	2.9	0.4	1.1	0.5	0.6	0.0	0.0	0.2	0.6	0.8	0.3	2.8
Sum non-target		773	26	573	7	87	194	28	75	36	43	0	0	13	37	50	17	187
%		100			0.9	11.3	25.1	3.6	9.7	4.7	5.6	0.0		1.7	4.8	6.5	2.3	24.2

Table 3
Correlation between performance on BNT in relation to demographic variables and cognitive function

	BNT performance					
	Responses			Ordinated responses		
	Correct	Omitted	Erroneous	Sub	Super	Side
Age	ns	ns	.22*	ns	ns	.19*
Gender	ns	.20*	ns	.29**	ns	ns
Education	.33***	−.25**	−.32***	ns	ns	−.30***
FSIQ	.40**	ns	−.28*	ns	ns	ns
Letter fluency	.29*	ns	ns	ns	ns	ns
Category fluency	.31*	ns	−.30*	ns	ns	ns
SLDT	.46***	ns	−.30*	ns	ns	ns
IPW	.52***	−.30*	.36**	ns	.31*	ns

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

tion between the subtest Information and number of correct responses ($r = .663$; $df = 50$; $p < .001$), while the association to the nonverbal subtest Digit Symbol ($p > .10$) was not significant. Lexical knowledge and ability was significantly associated to number of correct responses as demonstrated by verbal fluency (Letter fluency and Category fluency) and especially the two tests assessing passive lexical knowledge (SLDT and IPW).

The reliability, or internal consistency, in terms of number of correct responses was studied by calculating the split-half correlation (odd vs. even items) and Cronbach's α. Results demonstrated moderate coefficients (0.74 vs. 0.76).

3.2.2. Number of omitted responses

For the number of omitted responses ("don't know," category 2), there was a weak significant association with gender showing that women tended to reply with "don't know" more often than men did. Furthermore, individuals with a shorter education tended to admit that they did not know the correct response, while the individuals with a longer education tended to suggest a response although not totally correct. Age was not significantly related to "don't know" response. The association between global cognitive ability (FSIQ) and total number of correct responses was close to significant ($r = .255$; $df = 50$; $p = .068$) and the result on the subtest Information was especially associated with number of correct responses ($r = .536$; $df = 50$; $p < .001$). Other measures of lexical ability, Letter and Category fluency, was not significantly associated to number of omitted responses and not either the lexical decision test (SLDT) or the pronunciation knowledge test (IPW) indicated an association between omissions and knowledge of uncommon words.

3.2.3. Number of erroneous responses

The number of definitely erroneous responses (categories 6, 8a,b,c,d) were significantly associated with low

Table 4
Language ability for all subjects

	$M \pm SD$
Letter fluency	38.20 ± 16.31
Category fluency	25.17 ± 5.97
SLDT	163.1 ± 39.8
IPW	62.8 ± 8.3

education ($r = -.319$; $df = 109$; $p = .001$), low global cognitive ability ($r = -.282$; $df = 50$; $p = .043$), poor performance of category fluency ($r = -.297$; $df = 45$; $p = .043$), low results in the lexical decision test ($r = -.295$; $df = 58$; $p = .022$), and poor pronunciation knowledge as assessed by the IPW test ($r = -.359$; $df = 46$; $p = .012$). There was a significant association between the number of erroneous responses and increasing age ($r = .223$; $df = 109$; $p = .019$).

3.2.4. Language ability

The results on Letter and Category fluency as well as lexical decision (SLDT) and pronunciation knowledge (IPW) for all individuals are presented in Table 4, and the results are within normal limits according to values for various reference groups of healthy middle-aged individuals (Lezak, 1995).

3.2.5. Summary

The main pattern of results, concerning the effect of demographic factors on the amount of correct responses to the Swedish BNT, was that gender and age were of minor importance, whereas level of education was clearly important for the result on BNT. Long education had a significant positive relation to a high result on BNT and a negative relation to number of don't know-responses. Global cognitive ability (FSIQ) and lexical ability (Letter and Category fluency, SLDT, and IPW) had a significant positive relation to a high result on BNT, as measured by number of correct responses. The intercorrelations between tests of lexical ability (BNT, Letter and Category fluency, SLDT, IPW) were studied by means of a principal component analysis resulting in two factors accounting for 78% of the total variance. The analysis demonstrated that, BNT was primarily associated with lexical knowledge (SLDT and IPW) and that Letter and Category fluency represented a second factor interpreted as lexical access. The corresponding significant negative relation to number of don't know-responses was not found. The two measures of difficulty (1) number of correct responses and (2) number of omitted responses seem to measure different aspects of ability.

3.3. Analysis of response pattern

The various responses to all the 60 BNT items were analyzed with respect to frequency, to lexico-semantic properties and to how they correlated with demography and cognition. The most common response was the target response (88.4%) or words subordinated in relation to the target word (8.6%). Obviously it was typical for healthy adults with different background and cognitive ability to try to express them in a specific manner. When a problem in the search process occurred a typical solution of the naming problem was to suggest a sideordinated word (2.9% of all responses) or a superordinated word (1.3%). Totally 2.2% of the responses were associations outside the semantic category but mostly within the proper semantic field or context (categories 8b,c; 1.6%). In some cases associations seemed to depend on divergent interpretations of the picture (category 8a; 0.5%), or of problem with the proper word form (category 8d; 0.6%). Seemingly, a normal way to deal with a naming problem was to suggest something close to target and to remain within the semantic category. This was almost as common as to abstain to respond (2.8%). The share of descriptive responses was in total 1.9%. No association was found between amounts of descriptive utterances to demographic background but instead to poor verbal fluency.

A strong relation between total number of errors and number of categories used as substitutions was found ($r = .727$; $df = 109$; $p < .001$), see Fig. 2.

The response pattern varied among the 60 items with respect to homogeneity, accessibility to synonyms, and opportunity to use subordinated words. The frequencies for the various responses showed that the responses were very homogeneous concerning the common and well-known items. All the 111 individuals named seven items (*bed*, *scissors*, *comb*, *helicopter*, *snail*, *accordion* and *escalator*) in the same way by. Synonyms (as defined above in 2) were used for five items (*broom*, *hanger*, *bench*, *muzzle*, and *abacus*). Subordinated words were used for 30 items (*tree*, *pencil*, *house*, *flower*, *saw*, *broom*, *mushroom*, *hanger*, *camel*, *mask*, *pretzel*, *bench*, *racquet*, *volcano*, *dart*, *canoe*, *wreath*, *dominoes*, *cactus*, *knocker*, *pyramid*, *accordion*, *noose*, *latch*, *tripod*, *scroll*, *tongs*, *yoke*, *trellis*, and *palette*). In five items, a subordinated word was a more frequent choice than the target word (*racquet*, *dominoes*, *knocker*, *scroll*, and *tongs*), but only in the case of the items *racquet* and *knocker* one single variant of a subordinated word had a higher frequency than the target word. The occurrence of phonological errors was very low, 0.1% of all responses.

The different categories of substitutions or faulty responses to the 60 items were analyzed in relation to background data on age, gender, education, as well as lexical ability and general cognitive ability.

3.3.1. Number of subordinated responses (response category 4)

Age and years of education was not significantly associated with number of subordinated words. Also for global cognitive ability and various measures of lexical

ability there was no reliable association in relation to the number of subordinated responses. However, women used a significantly lower proportion of subordinated words ($r = -.319$; $df = 109$; $p = .001$).

3.3.2. Number of superordinated responses (response category 5)

Among the incorrect responses, the superordinated response was not very common (1.3%) in comparison with the other response types (see Table 2). Women used superordinated words to a significantly higher degree than men did ($r = .277$; $df = 109$; $p = .005$). Age or education was not important. A negative correlation was found between number of superordinated responses and high results on the verbal subtests in WAIS-R Information ($r = .412$; $df = 54$; $p = .002$) and Similarities ($r = -.511$; $df = 52$; $p < .001$). A high proportion of superordinated responses were also negatively associated with the pronunciation test (IPW; $r = -.309$; $df = 46$; $p = .033$), but it had no relation to verbal fluency (Letter and Category fluency).

3.3.3. Number of sideordinated responses (response category 6)

No effect was found for gender, but a higher age seemed to have some influence on the amount of sideordinated responses ($r = .191$; $df = 109$; $p = .043$), that is, they used another word in the semantic category, for instance *rat* instead of *beaver*. Also a shorter education was associated with a higher proportion of sideordinated responses ($r = -.300$; $df = 109$; $p = .001$). No reliable association was found in relation to the other measures of lexical ability or general cognition.

3.3.4. Number of associations outside the semantic category

A higher amount of visual associations (response category 8a) had some negative relation to education ($r = -.21$; $df = 109$; $p = .029$) and WAIS-R ($r = -.31$; $df = 52$; $p = .024$), while age and gender had no influence on visual associations. Lexical ability such as reading of uncommon words was related to number of visual errors ($r = -.381$; $df = 46$; $p = .037$). Phonological associations (response category 3) were related to a higher age ($r = -.244$; $df = 109$; $p = .009$) and to gender, in that men had a higher proportion of phonological associations ($r = .244$; $df = 109$; $p = .010$) than women. Phonological associations were also weakly related to performance on SLDT ($r = -.282$; $df = 56$; $p = .029$). Semantic or contextual associations were not related to either demographic or cognitive factors. When the five different types of associations (response categories 8a,b,c,d and 9) were added, no reliable effects of demographic or cognitive factors were found, whereas the relationship with lexical ability as exemplified by IPW pronunciation test was significant ($r = -.304$; $df = 46$; $p = .031$).

3.3.5. Specificity of descriptions

The total number of substituting descriptive utterances (response categories 10a,b,c,d) was not associated with age, gender or education. The amount of descriptive utterances was negatively associated to results on category fluency ($r = -.390$; $df = 45$; $p = .007$), verbal fluency ($r = -.260$; $df = 67$; $p = .31$), and lexical decision test ($r = -.319$; $df = 58$; $p = .013$). No association was found to the results on WAIS-R or IPW ($r = -.269$, $df = 46$; $p > .10$). The amount of responses that were more or less specific descriptions (response categories 10a,b,c,d) was not reliably associated to demographic or cognitive factors. This means that individuals with high education and high verbal intelligence (as tested with WAIS-R subtests Information, Similarities) gave as unspecific descriptions as those given by the individuals with low education and lower results on WAIS-R.

3.3.6. Summary

The main pattern of the analysis of naming responses was that the number of errors and number of categories used as substitutions was strongly associated. This means that the more difficult an item was, the more variations in the response pattern. Women used a higher amount of superordinated words and a lower amount of subordinated words, which means that their naming substitutions were less specific than the male participants. Specificity (low use of superordinated words) also had a significant positive relation to high results on the verbal subtests on WAIS-R. A higher total number of descriptive utterances were associated with low verbal ability.

4. Discussion

4.1. Standardization

In this paper, the adaptation of the original English version of BNT to a Swedish version is described. Responses by a sample of healthy adults to BNT are reported. The sample was smaller than in other similar studies on native populations (see Mariën et al., 1998), but on the other hand the sample is representative of the variation in age, gender, and length of education in the Swedish population. The subjects were tested with other lexical tests and half of the group also performed a short form of WAIS-R (Information, Digit Span, Similarities, Block Design, and Digit Symbol). The choice of language tests focused on lexical ability by assessing (1) *word retrieval* by verbal fluency tests (COWAT), (2) the *passive lexicon* by test of the ability to separate between words and nonwords (SLDT), and (3) knowledge of *unfamiliar words* by a pronunciation test (IPW).

The adaptation of the English version to a version suitable for the Swedish context and language was based on a response analysis of actual choices of lexical terms

made by the healthy population. The analysis showed that no items had to be replaced due to cultural differences, which was a satisfactory result. The investigation made it possible to suggest Swedish scoring rules and norms. The result of the response analysis was used to decide which Swedish translation that best corresponded to the typical naming of the pictures in Swedish. A preliminary selection of words, hitherto used in Swedish BNT-testing, could thereby be somewhat revised and a new version suggested. A definition of what should be scored as correct or incorrect was suggested.

4.2. Number of correct responses and cognition

The results from the investigation of the Swedish sample of healthy adults scored with the new norms showed that the frequency distribution of number of correct responses was close to the normal distribution. The mean score and standard deviation of correct responses, as found in the present study (53.04 ± 4.18), was comparable with international norms (Borod, Goodglass, & Kaplan, 1980) and the increasing difficulty among the items 1–60 was also comparable. The effect of age and gender on number of correct responses was small and not significant, while measures of education, general cognitive ability, and most prominently, a higher lexical knowledge and ability had a clear effect on number of correct responses. These results are mostly in line with what previously has been reported (Neils et al., 1995; Tombaugh & Hubley, 1997; Welch et al., 1996).

The results concerning the relation between BNT and other tests on verbal ability confirmed that BNT is a useful tool for assessments of naming and lexical ability.

When interpreting all the statistically significant results, it has to be kept in mind that there is always a risk that some results are spurious because of type I errors. Therefore, the pattern of results has to be considered rather than single significant results. In particular, this is important when correlations are based on relatively few data points.

The results of the study also raised the question if motivation to give a response has a relation to level of education. Omittances of responses were positively associated with low education, which supports the idea that the level of education has an effect on self-confidence and gives a different attitude against delivering some kind of response. Responses that were obviously wrong (sideordinated words and associations) constituted 5.5% of all the substitutive responses and this form of responses was associated with a shorter education and lower cognitive ability and lexical knowledge.

4.3. Comments on reliability

The reliability of the Swedish BNT was moderate, which has to be evaluated in relation to reliability of other internationally known versions of BNT as well as the original BNT. The internal consistency (Cronbach's α) of the original 60-item version of BNT has been reported to be 0.78 in healthy individuals (Tombaugh & Hubley, 1997) and 0.98 in aphasic patients (Goodglass & Kaplan, 1983). For healthy individuals, the Swedish BNT has a reliability that is in correspondence with similar version of BNT, both in English language and other languages. For patients with various kinds of language disturbance, the reliability is robust. However, it has to be kept in mind that BNT is intended as an instrument for assessment of clinically relevant difficulties rather than evaluation of performance in a healthy population.

4.4. Response analysis

Linguistic properties have previously been studied with BNT as a point of reference (see for example, Chenery, Murdoch, & Ingram, 1996, sometimes as part of standardization Tombaugh & Hubley, 1997). A similar approach was used in the present study and a semantic analysis of the various responses was performed. One purpose with the method and terminology used was to find a basis for further research with clinical samples. The present study used semantic concepts of attributiveness, closeness to target and different forms of associations. Classification into different forms of associations is not easy and no single solution exists. It is especially difficult to separate between semantic, contextual, and visual associations. One option could have been to use more classes. Descriptive utterance (categories 10a,b,c,d) was a term used for no dictionary terms or clauses that were used as substitutions for target responses. In other studies, descriptive utterances are named as circumlocutions, paraphrases or neologisms (Mariën et al., 1998). This taxonomy is appropriate and useful in clinical contexts but is not suitable for the analysis targeted in the present study. Unfortunately, the use of different types of taxonomy makes comparisons difficult and such a discussion could not be included in this study.

4.5. The naming process and the semantic field

The response analysis showed that healthy normal subjects respond close to target and mostly within the correct semantic fields also when their responses not can be scored as correct. The analysis also showed an expected strong relation between total number of various responses and amount of correct responses. The naming process is a function of a search process in the semantic network and it was interesting to notice how the search field was larger the more difficult the items were. The unfamiliar items caused attempts to name by use of a high amount of various categories accounted for above. One interpretation of this finding is that a higher

degree of difficulty in a naming task causes a more vivid and broad searching for some kind of substitutive utterance. But, the result of the search process does not only depend on the subject's familiarity with the items and the naming ability as discussed previously. Beside these factors also familiarity with the properties of the objects at hand have influence on the response pattern. Some aspects of this semantic issue will be discussed below and in relation to the results.

4.6. Semantic properties

During the work with the response analysis divergences were noted in the response pattern of single items, which might relate to semantic properties. One aspect of semantic properties of importance is the familiarity of the attributes that are associated with an item. Objects, like for example *noose* that have attributes that are familiar and easily named (like the attributive word *rope*) gave room for a high variety of responses, maybe because the semantic field was broad. Only one subject refrained from responding to the item *noose* while 30 subjects delivered some other response scored as incorrect. The opposite pattern was noted for the item *compass* that caused 20 omitted responses and 15 other various responses. The explanation might be that the two items *noose* and *compass* differ concerning quality of the semantic properties, although the level of difficulty was similar (*noose* 70% correct responses while *compass* got 78% correct responses). Another consideration concerning semantic properties is how easily and creatively an object can be described. In 1.9% of the total amount of responses, utterances classified as descriptive were used (categories 10a,b,c,d). The use of this form of substitution seemed to depend somewhat on the object in question and on the picture. Some items can be named by use of a short description or by creation of a portmanteau word while others do not share this opportunity. Conceptual complexity (*sphinx*) and richness of details in a picture (the picture of the *beaver*) might facilitate creation of descriptions and maybe also associations to functioning ("*posts which you walk upon*" for "*stilts*"). An example of an object with lower complexity and richness of details in BNT is "*asparagus*" with 86/111 correct responses but only one descriptive response. The object "*compass*" (86/111 correct responses) resulted on the contrary in 9 descriptive utterances. Maybe the examples of divergences mentioned above could motivate a separate study.

4.7. Comments on scoring

When it comes to clinical use, the scoring system of the present Swedish version might cause problems that are predictable. It can be difficult to decide for sure if a subordinated word should be scored as correct or not. Often, the use of subordinated words demonstrates a higher knowledge, like for example when naming the *racquet* a *tennis racquet*. In other occasions, the subordinated word might be wrong, for example if somebody says *spruce* for the picture of the *tree* (the picture seems to be a *broad-leaf tree*). A follow-up question could be of help in this scoring process as it is likely that normal adults know the superordinated *tree* (the target word). Tombaugh et al. suggested this procedure (1997). The use of the subordinated word possibly is due to an ambition or habit to be more specific. Some of the items might be so simple for normal adults that they cause embarrassment and lead to the choice of another and subordinated word. In the present study "incorrect" subordinated words like *dromedarian* for *camel* or *cottage* for *house* occurred in a few cases. To avoid arbitrariness, the words were scored as correct. Visual associations or misinterpretations occur now and then also in normal subjects naming and would probably be revealed by semantic cueing. Synonyms can sometimes be difficult to identify and the best way to handle this issue is to use a dictionary.

4.8. Concluding remarks

In summary, the typical way of responding to the naming test was to name the more familiar items in a very homogeneous way. It was also typical to respond with a subordinated word, which is to be more specific than necessary. The incorrect responses were usually close to the target concept and mostly within the correct semantic field. The share of correct responses had a strong relation to education, general cognition and above all to lexical ability. It was obvious that BNT not only measured naming ability but also indicated the level of lexical knowledge in general. The knowledge of the normal response patterns gives opportunities to perform comparative studies with clinical populations with lexico-semantic problems. The results are offered as norms for examination of native Swedish speaking adults.

Acknowledgments

Karolinska Institutet provided financial support for the present research. The subjects are thanked for kindly participating in the study. Speech Language Pathologists Lena Eriksson, Linda Henning, Ann-Kristin Nässström, and Klara Wenneborg are thanked for collecting data. Professor Ove Almkvist is thanked for help with the statistical analysis. Finally, PhD students and Speech Language Pathologists Ellika Schalling and Per Östberg are thanked for important discussions concerning the scoring method.

Appendix A

Target words of the English version and of the final version of the Swedish BNT

Item No.	English target words	Swedish target words	Item No.	English target words	Numb. correct
1.	bed	säng	1.	bed	111
2.	tree	träd	2.	tree	111
3.	pencil	penna	3.	pencil	111
4.	house	hus	4.	house	111
5.	whistle	visselpipa	6.	scissors	111
6.	scissors	sax	7.	comb	111
7.	comb	kam	9.	saw	111
8.	flower	blomma	11.	helicopter	111
9.	saw	såg	14.	mushroom	111
10.	toothbrush	tandborste	15.	hanger	111
11.	helicopter	helikopter	17.	camel	111
12.	broom	kvast	21.	racquet	111
13.	octopus	bläckfisk	22.	snail	111
14.	mushroom	svamp	28.	wreath	111
15.	hanger	galge	30.	harmonica	111
16.	wheelchair	rullstol	37.	escalator	111
17.	camel	kamel	47.	accordion	111
18.	mask	mask	8.	flower	110
19.	pretzel	kringla	10.	toothbrush	110
20.	bench	bänk	16.	wheelchair	110
21.	racquet	racket	18.	mask	110
22.	snail	snigel	19.	pretzel	110
23.	volcano	vulkan	25.	dart	110
24.	seahorse	sjöhäst	31.	rhinoceros	110
25.	dart	pil	46.	funnel	110
26.	canoe	kanot	5.	whistle	109
27.	globe	jordglob	12.	broom	109
28.	wreath	krans	36.	cactus	109
29.	beaver	bäver	43.	pyramid	109
30.	harmonica	munspel	27.	globe	107
31.	rhinoceros	noshörning	26.	canoe	106
32.	acorn	ekollon	33.	igloo	106
33.	igloo	iglo	34.	stilts	106
34.	stilts	styltor	39.	hammock	106
35.	dominoes	domino	20.	bench	104
36.	cactus	kaktus	13.	octopus	103
37.	escalator	rulltrappa	42.	stethoscope	101
38.	harp	harpa	29.	beaver	100
39.	hammock	hängmatta	23.	volcano	99
40.	knocker	dörrklapp	38.	harp	98
41.	pelican	pelikan	32.	acorn	98
42.	stethoscope	stetoskop	35.	dominoes	96
43.	pyramid	pyramid	54.	tongue	94
44.	muzzle	munkorg	24.	seahorse	93
45.	unicorn	enhörning	41.	pelican	93
46.	funnel	tratt	58.	palette	91
47.	accordion	dragspel	57.	trellis	90
48.	noose	snara	55.	sphinx	88
49.	asparagus	sparris	49.	asparagus	86
50.	compass	passare	50.	compass	86
51.	latch	regel	60.	abacus	85
52.	tripod	stativ	48.	noose	80
53.	scroll	rulle	59.	protractor	79
54.	tongs	tång	52.	tripod	74
55.	sphynx	sfinx	56.	yoke	74
56.	yoke	ok	44.	muzzle	67
57.	trellis	spalje	45.	unicorn	67
58.	palette	palett	53.	scroll	56
59.	protractor	gradskiva	51.	latch	51
60.	abacus	kulram	40.	knocker	39

References

Almkvist, O., Adveen, M., Henning, L., & Tallberg, I. M. (in preparation). Estimation of premorbid cognitive function based on lexical decision.

Almkvist, O., Bäckman, L., Basun, H., & Wahlund, L. O. (1993). Patterns of neuropsychological performance in Alzheimer's disease and vascular dementia. *Cortex, 29*, 661–673.

Benton, A. L., & Hamshire, K. de. S. (1989). *Multilingual aphasia examination*. Iowa City, Iowa, USA: University of Iowa Press.

Borod, J., Goodglass, H., & Kaplan, E. (1980). Normative Data on the Boston Diagnostic Aphasia Examination, Parietal Lobe Battery, and the Boston Naming Test. *Journal of Clinical Neuropsychology, 2*, 209–215.

Chenery, H. J., Murdoch, B., & Ingram, C. J. (1996). An investigation of confrontation naming performance in Alzheimer's dementia as a function of disease severity. *Aphasiology, 10*, 423–441.

Henderson, V. W., Mack, W., Freed, D. M., Kempler, D., & Andersen, E. S. (1990). Naming consistency in Alzheimer's disease. *Brain and Language, 39*(4), 530–538.

Goodglass, H., & Kaplan, E. (1983). *The assessment of aphasia and related disorders* (second ed.). Philadelphia: Lea and Febiger.

Hodges, J. R., Salmon, D., & Butters, N. (1991). The nature of the naming deficit in Alzheimer's and Huntington's Disease. *Brain, 114*, 1547–1558.

Kaplan, E., Goodglass, H., & Weintraub, S. (1983). *The Boston Naming Test*. Philadelphia: Lea and Febiger.

Kim, H., & Na, D. L. (1999). Normative data on the Korean version of the Boston Naming Test. *Journal of Clinical and Experimental Neuropsychology, 21*(1), 127–133.

Kohnert, K. J., Hernandez, A. E., & Bates, E. (1998). Bilingual performance on the Boston Naming Test: Preliminary norms in Spanish and English. *Brain and Language, 65*(3), 422–440.

Lezak, M. D. (1995). *Neuropsychological assessment* (3rd ed.). New York: Oxford University Press.

Lopez, M. N., Arias, G. P., Hunter, M. A., & Scott, R. R. (2003). Boston Naming Test: Problems with administration and scoring. *Psychological Reports, 92*(2), 468–472.

Margolin, D. I., Pate, D. S., Friedrich, F. J., & Elia, E. (1990). Dysnomia in dementia and in stroke patients: Different underlying cognitive deficits. *Journal of International Neuropsychological Society, 12*(4), 597–612.

Mariën, P., Mampaey, E., Vervaet, A., Saerens, J., & De Deyn, P. (1998). Normative data for the Boston Naming Test in native Dutch-Speaking Belgian elderly. *Brain and Language, 65*, 447–467.

Moberg, M., Ferraro, F. R., & Petros, T. V. (2000). Lexical properties of the Boston Naming Test stimuli: Age differences in word naming and lexical decision latency. *Applied Neuropsychology, 7*(3), 147–153.

Nationalencyklopedin (1997). Bokförlaget Bra Böcker AB, Höganäs.

Ord för ord, svenska synonymer och uttryck (1992). Norstedts Förlag AB.

Nelson, H. E., & O'Connell, A. (1978). Dementia: The estimation of premorbid intelligence levels using the New Adult Pronounciation test. *Cortex, 14*, 234–244.

Neils, J., Baris, J., Carter, C., Dell'aira, A., Nordloh, S., Weiler, E., et al. (1995). Effects of age, education, and living environment on Boston Naming Test performance. *Journal of Speech and Hearing Research, 38*, 1143–1149.

Nicholas, M., Obler, L. K., Au, R., & Albert, M. L. (1996). On the nature of naming errors in aging and Dementia: a study of semantic relatedness. *Brain and Language, 54*(2), 184–195.

Nordstedts Comprehensive English–Swedish Dictionary (1993). Norstedts Förlag, William Cloves Ltd.

Rubin, E. H., Strorandt, M., Miller, J. P., Kilscherf, D. A., Grant, E. A., Morris, J. C., et al. (1998). A prospective study of cognitive function and onset of dementia in cognitively healthy elders. *Archives of Neurology, 55*, 395–401.

Statistical Package for the Social Sciences (SPSS), version 10, 1999.

Sullivan, M., & Karlsson, J. (1998). SF-36 Hälsoenkät, Svensk manual och tolkningsguide. Sektionen för vårdforskning, Medicinska Fakulteten, Göteborgs Universitet och Sahlgrenska Sjukhuset.

Svenska Akademins ordlista över svenska språket (SAOL) (1998). Stockholm: Norstedts Ordbok, Svenska Akademin.

Tallberg, I. M., Wenneborg, K., & Almkvist, O. (in preparation). *Test of irregularly pronounced words and comparison with other lexical estimates of general cognitive function*.

Tombaugh, T. N., & Hubley, A. M. (1997). The 60-item Boston Naming Test: Norms for cognitively intact adults aged 25 to 88 years. *Journal of Clinical and Experimental Neuropsychology, 19*(6), 922–932.

Van Gorp, W. G., Satz, P., Kiersch, M. E., & Henry, R. (1986). Normative data on the Boston Naming Test for a group of normally older adults. *Journal of Clinical and Experimental Neuropsychology, 8*(6), 702–705.

Welch, L., Doineau, D., Johnson, S., & King, D. (1996). Educational and gender normative data for the Boston Naming Test in a group of older adults. *Brain and Language, 53*, 260–266.

Wechsler, D. (1981). *Manual for the Wechsler Adult Intelligence Scale—Revised*. New York Psychological Corporation.